

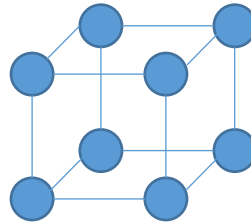
Planar Graphs

Planarity Testing

Planar Separator Theorem + Applications

Basics

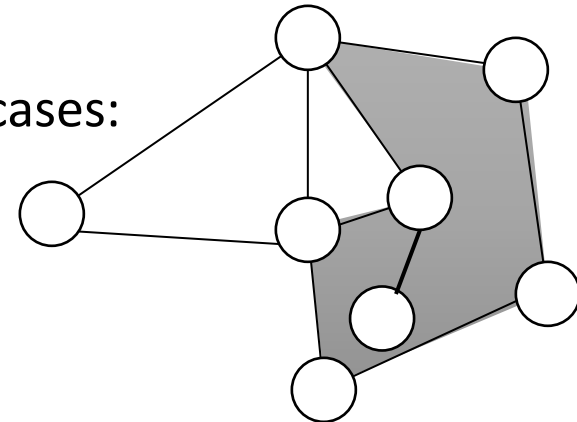
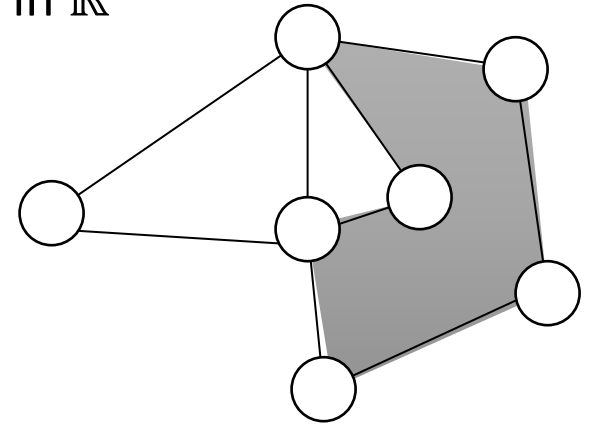
- Planar graph: Graph that can be drawn in $\mathbb{R}^2$ without edge crossings
- Plane graph: Planar graph with fixed embedding in $\mathbb{R}^2$ without edge crossings



- Is this graph planar?

- Faces, Outer face, Interior faces

- Each edge is incident to 2 faces, except in special cases:



Basics

- **Thm (Euler)**

Let G be a connected plane graph with $n \geq 1$ vertices, m edges, and f faces. Then $n + f = 2 + m$.

- **Pf** Induction on m .

- True if $m=0$
- If G has a cycle, then delete an edge and merge face
- If G has a vertex v of degree 1, then delete v and edge

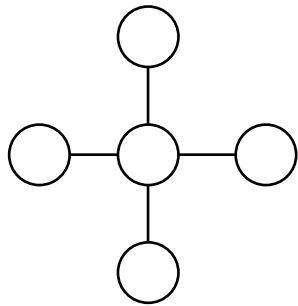
Basics

- If G is a connected plane graph with no parallel edges and no self-loops and $n > 2$, then $m \leq 3n - 6$
 - Every face 'has' at least three edge-sides, each edge-side 'is on' two faces, or twice on the same face
$$3f \leq 2m$$
$$m = n + f - 2 \leq n + 2m/3 - 2$$
$$m/3 \leq n - 2$$
- Is K_5 planar?
- Every plane graph with no parallel edges and no self-loops has a vertex of degree at most 5
 - Average degree is $2m/n \leq 2(3n - 6)/n < 6$

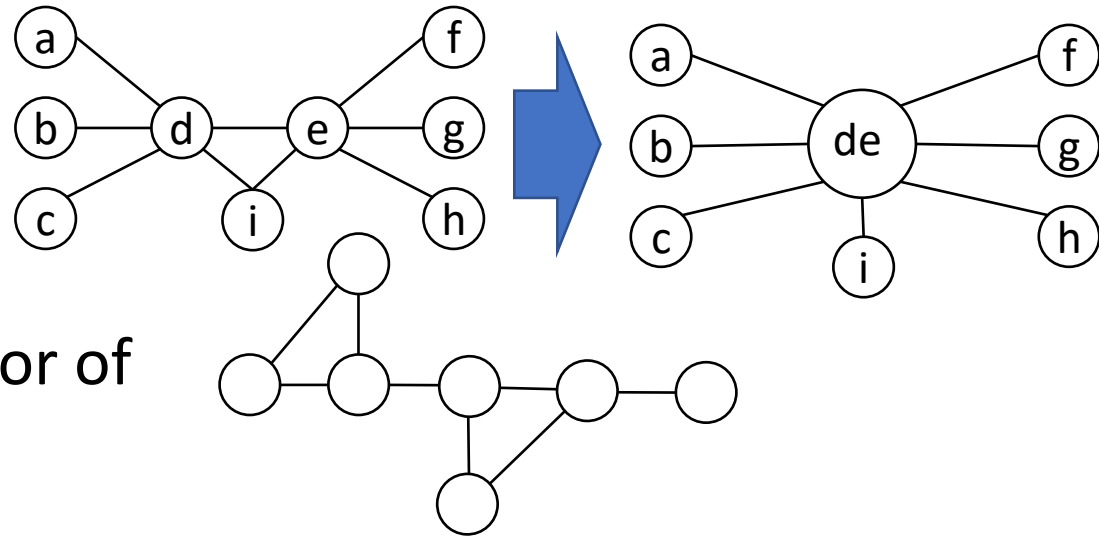
Basics

- H is a minor of G : H can be obtained from G with edge deletion, vertex deletion and edge contraction

- Edge contraction:



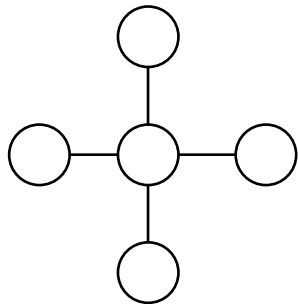
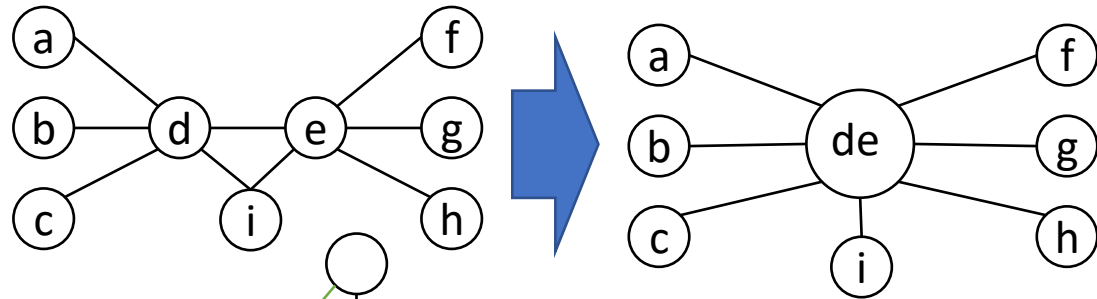
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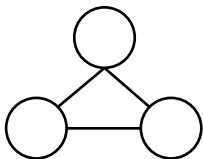
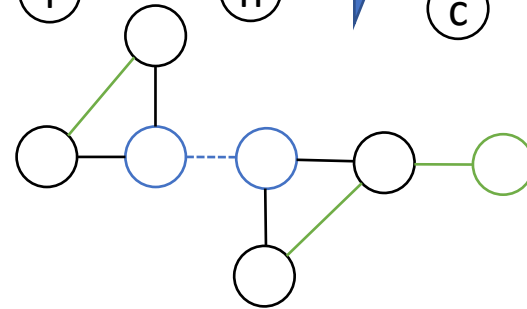
Basics

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is a minor of

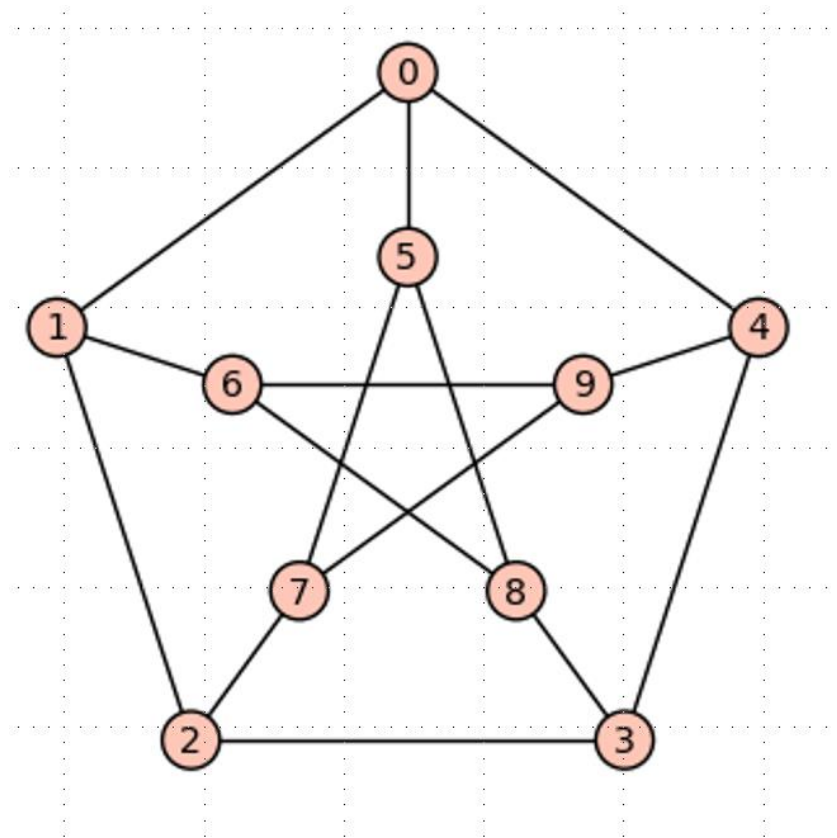


is a minor of G iff G has a cycle

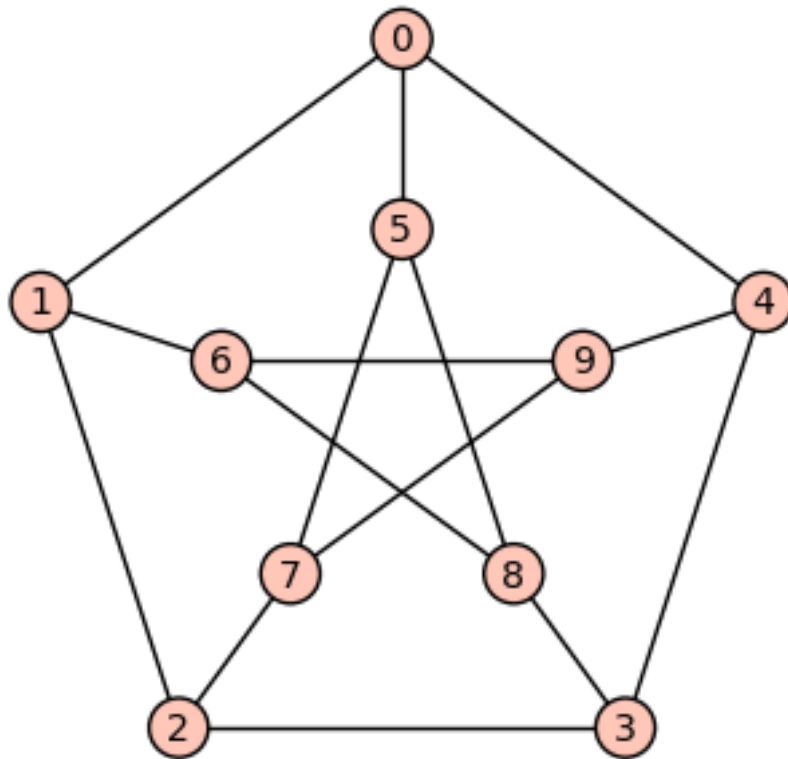
- **Thm(Kuratowski / Wagner):**

G planar iff it has no K_5 or $K_{3,3}$ as minor

Basics : planar or not?



Basics : planar or not?



Basics

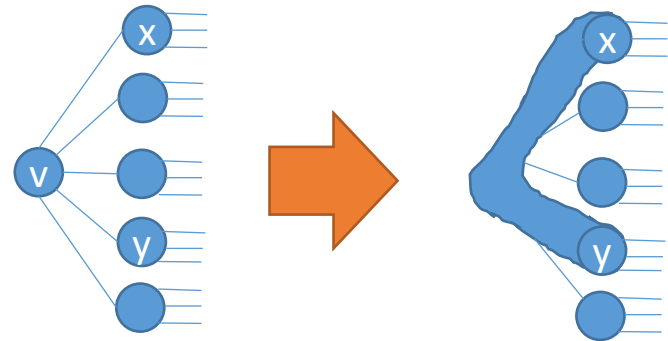
Recall: q -coloring is $c: V \rightarrow \{1, \dots, q\}$ s.t. $c(v) \neq c(w) \forall \{v, w\} \in E$

Planar graphs are 6-colorable

- Induction on $|V|$, remove vertex of degree ≤ 5

Planar graphs are 5-colorable:

- Induction on $|V|$:
- Consider $v \in V$ of degree ≤ 5
 - If $d(v) < 5$, use induction as before
 - If $d(v) = 5$, let x, y be non-adjacent neighbors (by Kuratowski!)
 - Use induction on G' obtained by removing v and unifying x, y
 - G' is planar since it is a minor of G (remove $\{v, w\}$ for $w \neq x, y$ and contract $\{v, x\}, \{v, y\}$)



Big Theorem: Planar graphs are 4-colorable

Complicated proof, need computer for huge case analysis

Basics

Theorem (Fary, 1936): Any planar graph can be
drawn such that edges are straight lines

There is a whole research area called *graph drawing*

- Want to draw (almost) planar graphs in various ways with various desirable properties

Planarity Testing

Linear time algorithms known.

We focus on an insightful $O(n^3)$ (polynomial) time algorithm

Cannot use Kuratowski's theorem for polynomial time algo.

Assume connected

Assume 2-connected (each vertex lies in a cycle)

[If removing v disconnects, test components separately]

Key insight:

Consider a cycle C . If $\{u, v\}$ an edge with u and v not on C , then either both u and v lie inside C or both lie outside C .

Planarity Testing

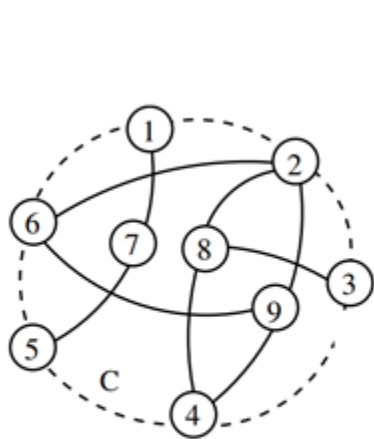
Fix a cycle C . Induces a set of *segments* S_i :

- S_i either is an edge that connects two non-consecutive vx's of C or
- S_i is induced by the edges of a connected component of $G - V(C)$ together with the edges connecting that component to C .

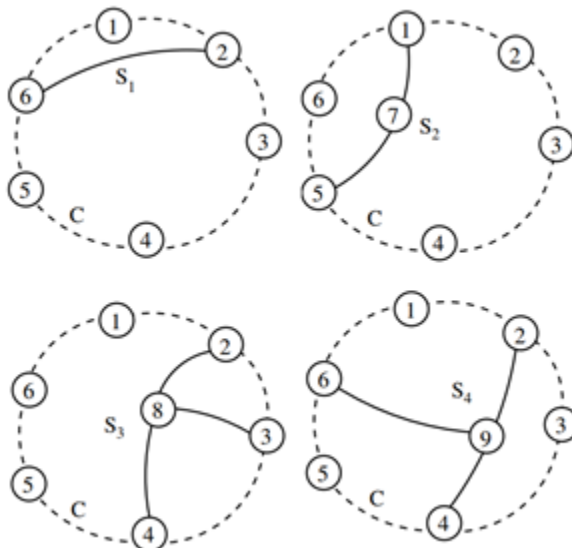
Conflicting segments: Cannot be drawn on same side of cycle:

- More formally, no c_1, c_2, c_3, c_4 in cycle in this order with $c_1, c_3 \in S_i, c_2, c_4 \in S_j$

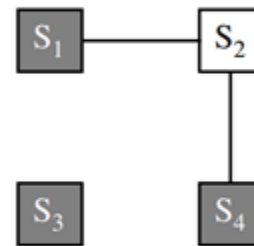
Interlacement graph H : vx for each segment, edge if segments 'conflict'



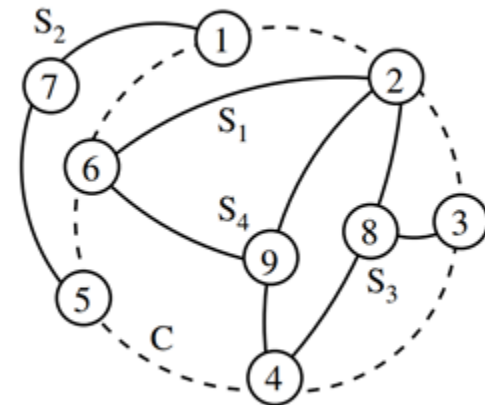
G



Segments



Interlacement
Graph



Drawing

Algorithm outline

- Find cycle C , and let $S_1, \dots, S_k$ be segments.
- If interlacement graph not 2-colorable
 - return NO
- Else
 - recursively test planarity of graph $C + S_i$, for each $i = 1, \dots, k$

If interlacement graph not 2-colorable or some $C + S_i$ not planar,
 G clearly not planar

Otherwise, merge drawings of $C + S_i$ for $i = 1, \dots, k$
2-coloring indicates the segments inside the cycle

Almost there, except one catch

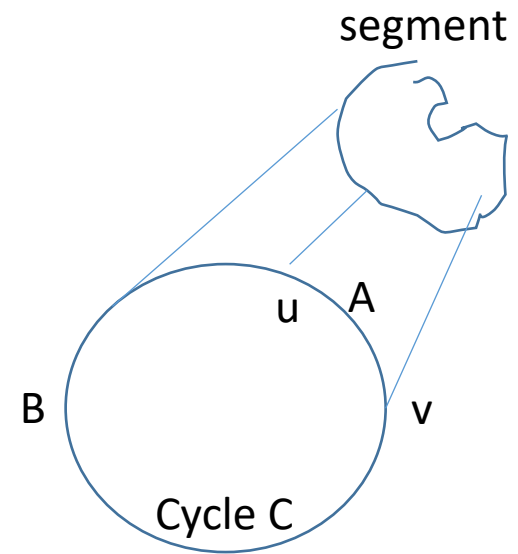
What if $C + S_1 = G$?

Def: Call a cycle **separating**, if has ≥ 2 segments.

Lem: G has a separating cycle C unless,

- (i) G is itself cycle or,
- (ii) G is a cycle + segment which is a path

Moreover can find C in polynomial (linear) time.



A, B are the two paths from u to v contained in C

Pf Find some C . If it has ≥ 2 segments or one path segment, done

- Else it has one segment S . Take two consecutive attachments u and v (u, v exist by 2-connectivity; S may have more than 2 attachments)
- Take some path P from u to v in the segment S (not involving C)
- Let $C' := P + B$, where B is the path on C from u to v that contains the other attachments of S (if any)
 - Unless $S = P$, C' separates other attachments or other parts of the segment from A

Algorithm outline

- Find cycle **separating** C , and let $S_1, \dots, S_k$ be segments.
- If interlacement graph not **2-colorable**
 - return NO
- Else
 - **recursively** test planarity of graph $C + S_i$, for each $i = 1, \dots, k$

If interlacement graph not 2-colorable or some $C + S_i$ not planar,
 G clearly not planar

Otherwise, merge drawings of $C + S_i$ for $i = 1, \dots, k$
2-coloring indicates the segments inside the cycle

Number of recursive calls is at most m , by a tricky argument that assigns a unique edge to each recursive call (details omitted)

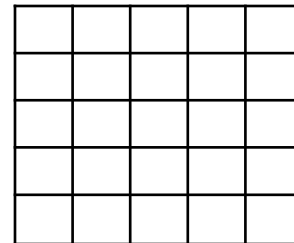
Planar Separator Theorem

Thm: Let $G = (V, E)$ be a planar n -vertex graph. There is a $9/10$ -balanced separator of size $11\sqrt{n}$.

- Several proofs exist. Can also find the separator in poly time.
- Optional: Geometric proof on blackboard. Skipped in lecture

Thm tight up to constant factors:

No $9/10$ -balanced sep. of size $n/20$ in the $(n \times n)$ -grid $G = (V, E)$.



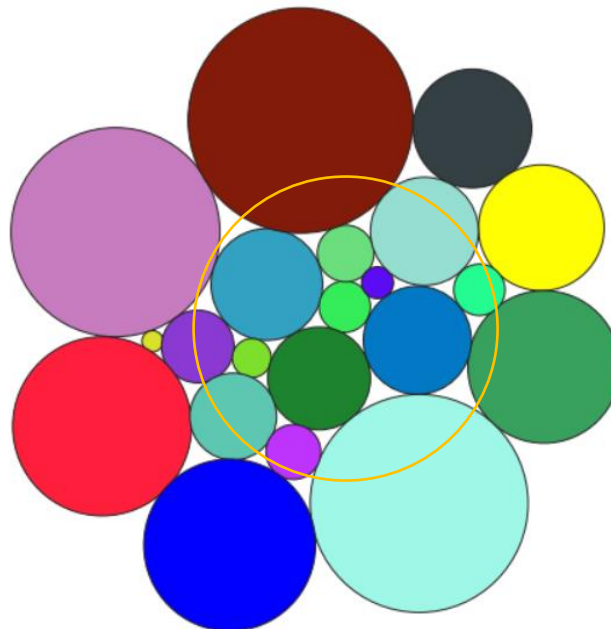
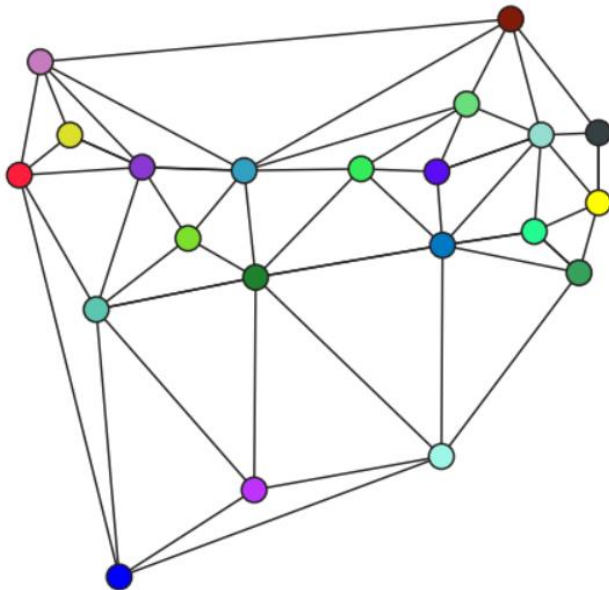
Idea Proof Planar Separator Theorem (PST)

Koebe's disk embedding theorem:

Can draw a planar graph G such that for each vertex v , there is a disc D_v , with the following properties:

- (i) Interiors of disks are disjoint
- (ii) two discs D_v and D_w touch iff $\{v,w\}$ is an edge in G .

Geometric proof of pst via *probabilistic method*



Proof Sketch of Planar Separator Theorem

Consider disk embedding of G

Let D be the smallest circle that contains $\geq n/10$ center points

Assume D is of radius 1 and centered at origin

Pick x uniformly at random in interval $[1,2]$

Consider circle C_x centered at origin of radius x

Let $S = \{v \in V : D_v \text{ that intersects } C\}$

(note S is a random set)

S will be small balanced separator with good probability
(omitted, optional reading)

Application Planar Separator Theorem

Thm: Max. Independent set on planar graphs in $2^{O(\sqrt{n})}$ time

- Set $OPT = \emptyset$
- Let S be separator, $V_1, \dots, V_k$ be connected components of $G[V \setminus S]$
- For every independent set X of $G[S]$
 - Recursively find max IS OPT_i of $G[V_i \setminus N(X)]$ for $i = 1, \dots, k$
 - If $|X \cup_i OPT_i| > OPT$, replace OPT with $X \cup_i OPT_i$
- Output OPT

Run time recurrence: Let $T(n)$ be run time in terms of $|V|$

$$T(|V|) \leq |V|^{O(1)} 2^{|S|} (\sum_i T(|V_i|))$$

$T(n) \leq 2^{1000\sqrt{n}}$ fits for n large enough:

$$T(n) \leq n^{O(1)} 2^{11\sqrt{n}} T(9n/10) \leq 2^{11\sqrt{n} + 1000\sqrt{9n/10}} \leq 2^{1000\sqrt{n}}$$

Computational Problems on planar graphs

- Given the positive result, you might wonder: are NP-hard problems on planar graphs still NP-hard
- Most problems remain NP-hard on planar graphs
- One (of the several) notable exceptions:
Planar Clique: Given: Planar graph G , integer k
Asked: Does G have a clique of size at least k ?
- How can you solve **Planar Clique** in poly time?
- As a sampler, we'll now show Planar 3-Coloring to be NP-hard by a reduction from 3-Coloring
- **(Planar) 3-Coloring: Given:** (Planar) graph G
Asked: Does G have a 3-coloring?

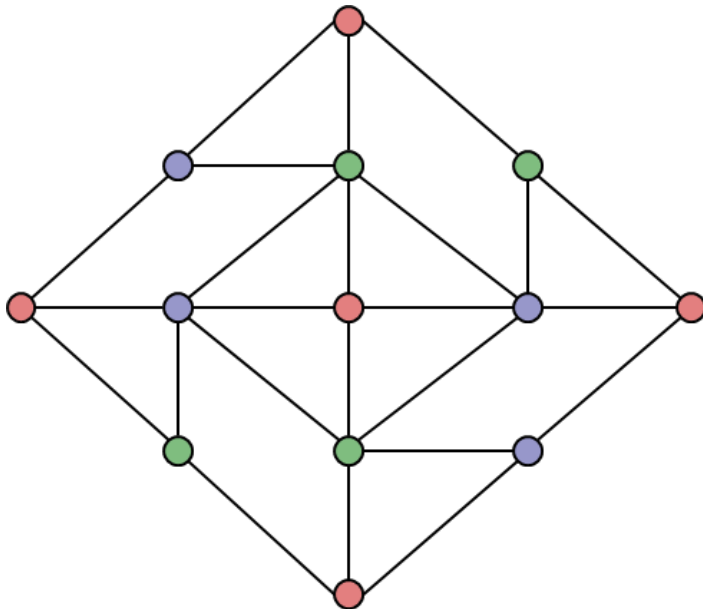
Planar Graph 3-Colorability

Claim. There is a planar graph W such that:

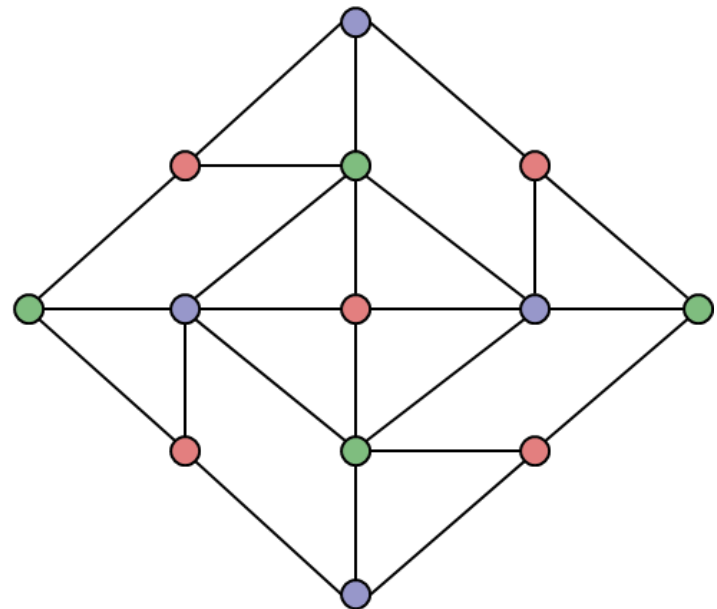
- In any 3-coloring of W , opposite corners have the same color.
- Any assignment of colors to the corners in which opposite corners have the same color extends to a 3-coloring of W

Pf. W has only two 3-colorings (or by permuting colors).

four corners same color



four corners two colors

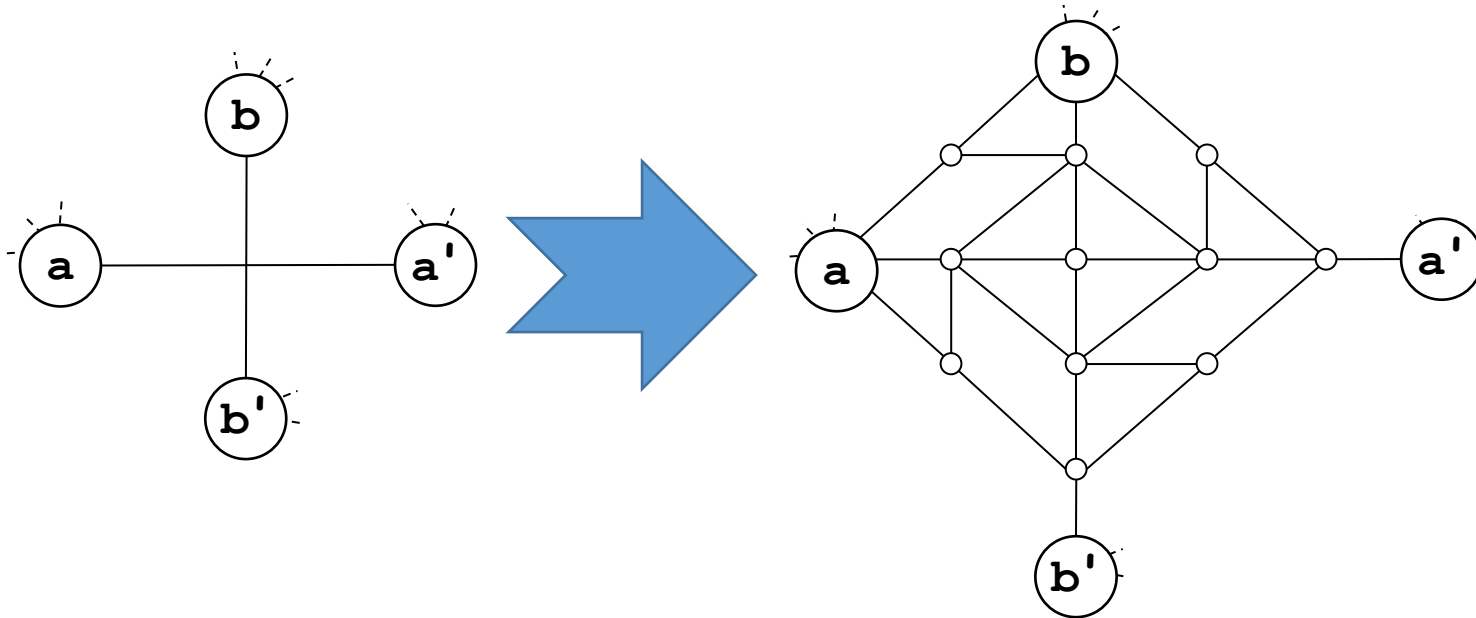


Planar Graph 3-Colorability

Claim. $3\text{-COLOR} \leq_p \text{PLANAR-GRAPH-3-COLOR}$

Pf. Given instance of 3-COLOR, draw graph in plane, letting edges cross

- Replace each edge crossing with planar gadget **W**
- In any 3-coloring of **W**, $a \neq a'$ and $b \neq b'$
- If $a \neq a'$ and $b \neq b'$ then can extend to a 3-coloring of **W**



a crossing

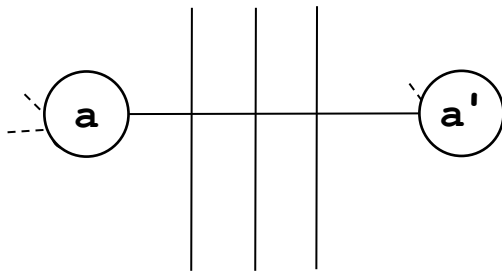
gadget **W**

Planar Graph 3-Colorability

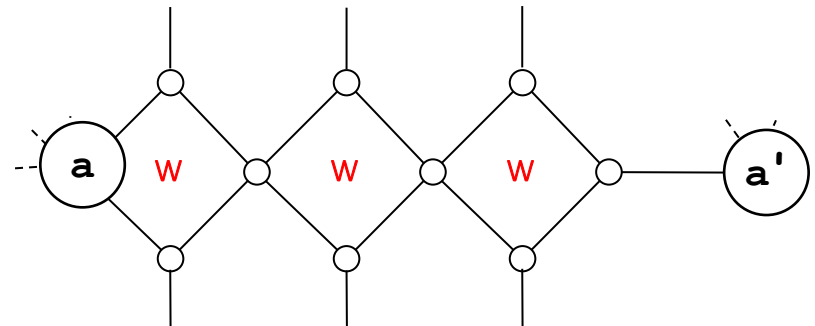
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Pf. Given instance of 3-COLOR, draw graph in plane, letting edges cross

- Replace each edge crossing with planar gadget W
- In any 3-coloring of W , $a \neq a'$ and $b \neq b'$
- If $a \neq a'$ and $b \neq b'$ then can extend to a 3-coloring of W



multiple crossings



gadget W